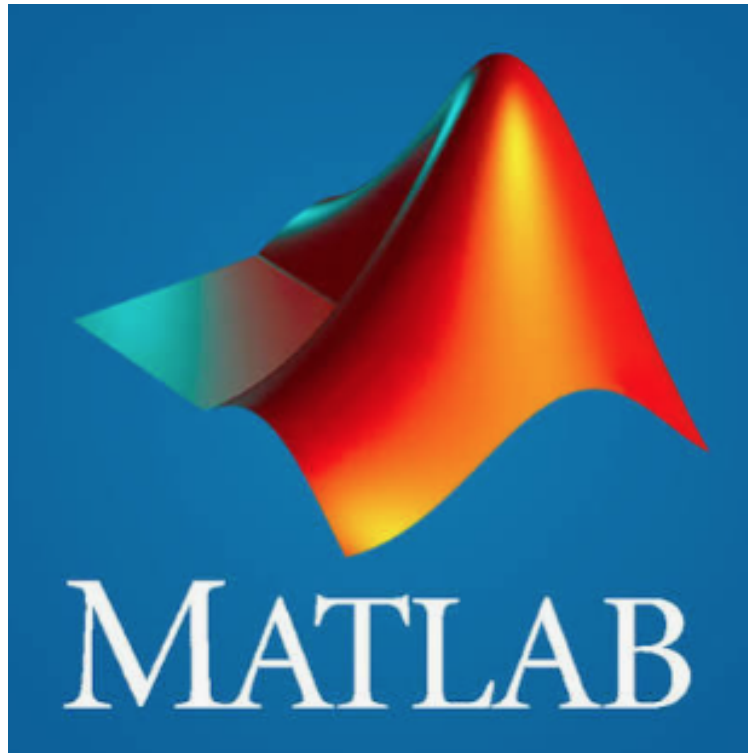
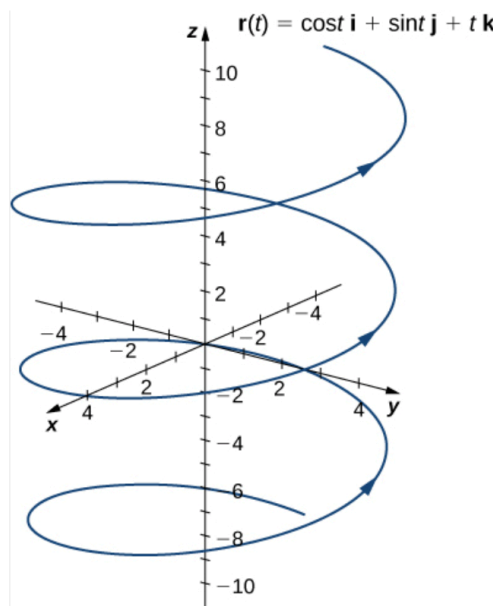


MATH 280 MATLAB Project #2 - Vector Motion and Trajectory Simulator



When the project is completed, rename the end of the file to (Completed Implementation).



PROJECT OUTLINE

Core Objective

Develop a MATLAB application that models and visualizes motion in three-dimensional space using vector-valued functions. The application allows users to create trajectories, simulate projectile motion, compute arc length, analyze velocity and acceleration vectors, and examine tangent and normal vectors along a path through an interactive GUI.

Rather than acting as a simple graphing program, the application serves as an interactive 3D Motion and Trajectory Analysis Tool.

Concepts Covered

Chapter 13 Vector-Valued Functions in Motion & Space

- 13.1 Vector Functions
- 13.2 Modeling Projectile Motion (physics app)
- 13.3 Arc Length and the Unit Tangent Vector T
- 13.4 Curvature and the Unit Normal Vector N

Learning Objectives

By completing this project you will practice:

1. vector-valued functions
2. parametric motion
3. numerical differentiation and integration
4. physics modeling
5. interactive visualization
6. engineering analysis
7. MATLAB GUI development

Required Features

1. Vector Function Creator in MATLAB that plots:

- trajectory
 - direction arrows
 - starting point
 - ending point
2. Projectile Motion Simulator in MATLAB that computes:
- trajectory
 - maximum height
 - range
 - total flight time

And displays the animated projectile motion in 3D.

3. Velocity Visualization in MATLAB that computes derivative of $r(t)$ or $v(t)$ and displays:
- velocity vectors
 - speed graph
 - instantaneous velocity
4. Acceleration Visualization in MATLAB that computes derivative of $v(t)$ or $a(t)$ and displays:
- acceleration vectors
 - acceleration magnitude
 - velocity and acceleration together
5. Arc Length Calculator in MATLAB where the user selects a start and end time (limits of integration) and the program computes the arc length which is the definite integral of $v(t)$ and displays total distance traveled highlighted portion of trajectory.
6. Tangent Vector Analysis in MATLAB where the program computes the unit tangent vector $T(t)$ and displays:
- tangent vector
 - moving tangent arrow
 - tangent direction animation
7. Normal Vector Analysis in MATLAB where the program computes the unit normal vector $N(t)$ and displays:

- normal vector
- tangent vector
- comparison plot

8. Curvature Explorer in MATLAB where the program computes the unit curvature vector $\kappa(t)$ and displays:

- curvature versus time
- color-coded trajectory
- highest curvature points

$\kappa = \text{kappa}$

9. Motion Animation in MATLAB

Animate the following in MATLAB:

- moving particle
- velocity vector
- acceleration vector
- tangent vector
- normal vector

Animation controls/buttons:

- Play
- Pause
- Restart
- Speed Slider

10. Motion Comparison

Allow the user to compare two trajectories simultaneously in MATLAB

Example: Projectile vs Helix

Necessary MATLAB Files For This Project/File Structure

MVC_Project2/
main.mlapp
main.m
trajectoryGenerator.m
projectileMotion.m
velocityCalculator.m
accelerationCalculator.m
arcLengthCalculator.m
tangentVector.m
normalVector.m
curvatureCalculator.m
plotUtilities.m
[README.md](#) (for GitHub)

Useful MATLAB Commands For This Project

plot3()
animatedline()
comet3()
quiver3()
gradient()
integral()
trapz()
norm()
diff()
meshgrid()
hold on
axis equal
view()
uifigure()
uiaxes()

uibutton()
uislider()
uidropdown()
uicontrol()
uitable()

Additional MATLAB Pseudocode

Trajectory Generation

Receive vector function
Generate parameter values
Compute x, y, z
Plot trajectory

Projectile Motion

Receive launch conditions
Compute equations of motion
Generate trajectory
Animate projectile

Arc Length

Receive trajectory
Differentiate
Compute speed
Numerically integrate
Display distance

Tangent Vector

Receive trajectory
Differentiate

Normalize velocity

Display moving tangent vector

Curvature

Receive trajectory

Compute derivatives

Evaluate curvature

Plot curvature

Highlight maximum curvature

GUI FUNCTIONALITY

1. Function input field for entering vector-valued functions
2. Drop-down menu of predefined trajectories (helix, circle, ellipse, spiral, projectile, Lissajous curve)
3. Interactive sliders for animation speed and simulation time
4. Checkboxes to show or hide certain parts
5. Velocity vectors
6. Acceleration vectors
7. Tangent vectors
8. Normal vectors
9. Live numerical panel updating position, speed, acceleration, arc length, and curvature
10. Export graph as a downloadable image
11. Reset simulation button

MAIN MENU

Vector Motion Simulator

1. Create Vector Function

2. Simulate Projectile Motion

3. Plot Velocity

4. Plot Acceleration

5. Compute Arc Length

6. Analyze Tangent Vector

7. Analyze Normal Vector

8. Curvature Analysis

9. Animation

10. Exit

GUI LAYOUT

Primary Interface

Left Panel

- **Function input**
- **Time interval**
- **Initial velocity**
- **Launch angle**
- **Gravity selection**
- **Animation speed**

Center

Large interactive 3D plot

Right Panel

Displays:

- **Position**
- **Velocity**
- **Acceleration**
- **Arc length**
- **Curvature**
- **Current time**

Bottom

Buttons:

- **Plot**
- **Animate**
- **Pause**
- **Reset**
- **Export Figure**

FINAL DELIVERABLES

- **MATLAB App Designer project (.mlapp) providing the primary GUI.**
- **MATLAB source files organized into reusable functions for motion analysis and visualization.**
- **A README explaining the mathematical concepts, program architecture, GUI features, and instructions for running the application.**

- Sample demonstrations of vector-valued functions, projectile motion, tangent and normal vectors, curvature analysis, and animated trajectories.

multivariable calculus

$$\iiint f(x,y,z) \frac{\partial f}{\partial y} \frac{\partial f}{\partial x} 7i + 4j + 3z$$